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Axial Position Sensorless Detection and Control Method for 3-DOF Permanent-Magnet Synchronous Machine in Maglev Pump

Chaoning Zhu, Liang Hu, Xiaodong Ruan, Rui Su, Jinhua Chen, Chi Zhang, and Xin Fu

***Abstract*—The axial stability of a magnetically levitated pump is crucial for its operation. Conventional active control relies on physical sensors for axial position feedback, which are often impractical due to spatial and environmental constraints. To address this challenge, this article proposes a sensorless detection and control scheme that combines a novel position estimation method with a sliding mode controller (SMC). In this method, a variable forgetting factor recursive least squares algorithm is developed to accurately identify the flux linkage across a wide rotational speed range. The identified rotor position, derived from a linear relationship with the flux linkage, is then fed into an integral SMC designed to suppress uncertainties and estimation errors, thereby ensuring stability and eliminating steady-state error. Simulation and experimental results validate the high linearity and accuracy of the proposed method, demonstrating effective axial position control under disturbances.**

***Index Terms*—Axial position estimation, maglev pump, multidegree-of-freedom (DOF) suspension, online identification, rotor flux linkage, sliding-mode control (SMC).**

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I. Introduction

MAGNETICALLY levitated (maglev) pumps are fluid power elements that operate without lubrication, dynamic seals, or mechanical friction. Their suspension and driving systems enable levitation and rotation of the impeller [1]. Recent design advances have significantly improved both suspension and driving performance [2], [3], [4]. Based on the active suspension type, maglev pumps can be classified into axial 1-DOF, tilt and axial 3-DOF, 5-DOF, and radial 2-DOF suspension systems [5], [6], [7], [8]. Among these, radial 2-DOF suspension is the most widely used. However, impellers in 2-DOF active suspension systems may experience significant axial hydrostatic pressure, leading to axial displacement and potential collision with the pump casing [9], [10]. Dynamic reaction forces can induce axial impact [11], causing vibration or even instability [12]. To address this, a 3-DOF active suspension system with axial magnetic support was previously proposed, which enhances axial control accuracy and stability, allowing the system to withstand even more severe axial hydraulic forces [13].

The implementation of active axial control introduces two main challenges: the need for axial position sensing and the fact that system parameters vary with rotor displacement. Adding displacement sensors is often infeasible due to space constraints. Eddy current sensors require large coils relative to their measurement range [14], while Hall sensors are affected by magnetic interference from control currents [15], [16]. Magnetoresistive and resolver sensors face similar limitations [17]. Thus, noncontact sensing in enclosed systems remains challenging.

A promising sensorless approach is to estimate displacement by monitoring changes in system parameters, thereby eliminating the need for dedicated sensors. Existing methods include flux linkage model-based approaches [18], [19] and synchronous inductance model-based approaches [20], [21]. For example, in radial motors, the drive winding flux linkage has been used to estimate axial position [12], but its operating range is limited to one side of the stator. In axial flux motors, changes in synchronous inductance have been employed to estimate axial position [21]. However, the relationship between inductance and displacement is typically nonlinear.

In the area of rotor position estimation algorithms, several mainstream sensorless control methods have emerged in conventional systems. For instance, high-frequency injection methods enable operation at zero and low speeds [22], but suffer

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from torque ripple and audible noise. Model reference adaptive systems are structurally simple and easy to implement [23], yet they are sensitive to motor parameter drift. Extended Kalman filters offer strong noise rejection and full-speed-range adaptability [18], but they involve high computational complexity and require tuning of numerous parameters. Sliding mode observers are robust and simple to implement [24], though they exhibit inherent chattering and steady-state errors. Therefore, a sensorless solution that does not rely on a rotor angle model, can adapt online to parameter variations, and has moderate computational requirements is needed.

Among candidate methods, the recursive least squares (RLS) algorithm with a forgetting factor stands out as particularly suitable for applications with low computational cost requirements due to its moderate computational load [25]. Nevertheless, its conventional fixed forgetting factor poses a fundamental tradeoff: noise suppression at low speeds conflicts with rapid tracking at high speeds when speed varies significantly. While adaptive forgetting factors have been explored to address this issue [26], [27], their performance still falls short of the real-time requirements imposed by our maglev pump application. To this end, a variable forgetting factor RLS (VFF-RLS) method can be devised, aiming to balance the aforementioned conflicting requirements while meeting real-time constraints.

In the field of multidegree-of-freedom (DOF) active suspension control for magnetic levitation pumps, there is limited research on axial suspension control due to the distinct electromagnetic topologies employed. Currently, most systems employ proportional-integral-derivative (PID) controllers to achieve levitation due to their simplicity [28]. In this system, the force-current coefficient varies with the rotor's axial displacement, which can alter the system poles when a PID controller is used [13]. This necessitates adjusting the controller parameters to ensure the rotor's control performance. For systems susceptible to parameter and load variations, robust control methods are typically employed, such as sliding mode control (SMC) [29]. SMC is an effective approach for handling uncertainties and disturbances and has found wide application in the control of permanent magnet synchronous motors [30].

In maglev pumps with multi-DOF suspension, axial control remains underexplored. PID controllers are commonly used for their simplicity [28], but variations in system parameters caused by changes in axial displacement can alter the force-current coefficient and potentially shift the system's poles, necessitating controller retuning [13]. For systems prone to such uncertainties and disturbances, robust control methods like SMC are advantageous [29], [30]. Various SMC techniques have been developed, including adaptive, terminal, integral, and high-order SMC [31]. The choice of sliding surface is crucial for controller performance. For example, proportional–integral surfaces can eliminate steady-state error [32], [33], integral SMC can avoid the traditional reaching phase [34], and supertwisting algorithms enhance performance for second-order systems [35].

This article presents a practical system-level solution for sensorless axial control of a 3-DOF maglev pump. The main contribution is the innovative integration of two key components. 1) A novel VFF-RLS algorithm that dynamically adjusts to speed changes, enabling reliable position estimation across the entire speed range. 2) An SMC designed to ensure robustness against system uncertainties and observation errors. This integrated approach enables stable axial active control without adding any physical sensors, achieving significant space savings. The proposed method is applicable to any electromagnetic system with a PM-dominant field, axial-displacement-sensitive flux path, and linear response, e.g., axial magnetic bearings, axial flux motors, and magnetic couplings. The VFF-RLS algorithm is portable, computationally efficient, easy to tune, and enables wide-speed-range dynamic identification.

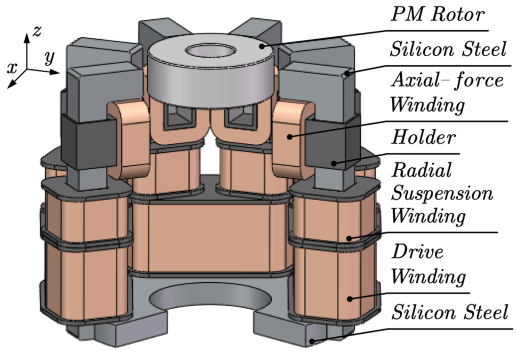


Fig. 1. Structure of the 3-DOF active suspension electromagnetic system for maglev pump.

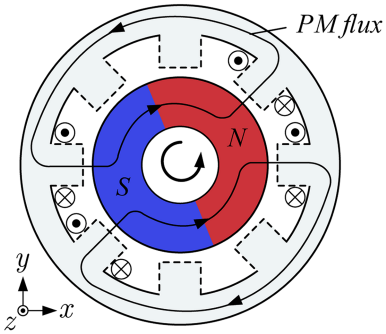


Fig. 2. Connection mode of axial-force winding.

The rest of this article is organized as follows. Section II describes the topology and operating principles. Section III presents the axial position estimation and control method. Section IV validates the flux linkage variation via simulation. Section V discusses experimental methods and results. Finally, Section VI concludes this article.

II. Topology and Operation Principle

The structure of the 3-DOF active suspension electromagnetic system is depicted in Fig. 1. The system features two-phase monopole-pair coreless axial-force windings positioned near the side surface of the one-pole pair permanent-magnetic rotor. Fig. 2 illustrates the connection mode in one phase. Further details regarding the system structure can be found in [13]. The system provides active suspension in 3-DOF, with axial, radial, and rotational motions controlled approximately independently.

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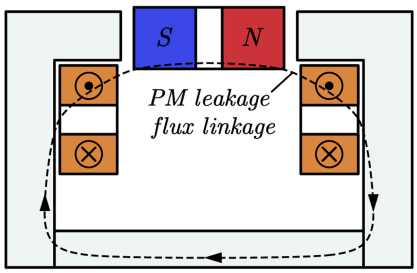


Fig. 3. Flux linkage of permanent magnet in axial-force winding.

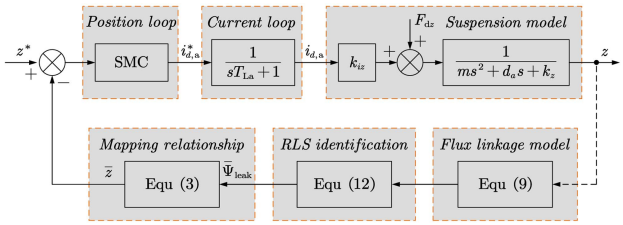


Fig. 4. Flowchart of sensorless axial levitation control strategy.

The radial suspension force is achieved by altering the air-gap flux density through the suspension winding. Then radial displacement signals are utilized as feedback to achieve suspension. The rotation of the rotor is accomplished by generating tangential flux on the rotor surface through the drive winding, which attracts the rotor to rotate. Angular position and velocity signals are then used as feedback to achieve rotational drive.

The active axial force is generated by producing magnetic flux and current in the axial winding that rotate synchronously with the rotor. This results in a constant Maxwell and Lorentz stress, which remains unchanged even as the rotor angle varies. However, the force magnitude varies with the axial gap between the winding and the rotor, causing the force-current coefficient to change with rotor displacement and introducing variations in system parameters.

As the rotor moves axially, the winding leakage flux changes correspondingly, which is shown in Fig. 3. Within the operating range, the rotor leakage flux linkage and axial displacement exhibit a linear relationship, allowing accurate axial position estimation through flux linkage detection.

III. Analysis of Position Estimation and Control Models

The schematic diagram of the sensorless control of the axial suspension system is shown in Fig. 4, including the flux linkage model, current loop model, RLS identification algorithm, axial suspension model, and sliding mode position controller. Key equations are annotated in the diagram. The models are described in sequence in the respective sections.

A. Analysis of the Mathematical Model of Flux Linkage in Permanent-Magnet Synchronous Machines

The radial and axial magnetic flux vectors on the surface of the permanent-magnet rotor exhibit a sinusoidal distribution due to parallel diameter magnetization. Consequently, the composite magnetic flux vectors at the axial-force winding also follow a sinusoidal pattern. As shown in Fig. 3, this flux component can be defined as rotor leakage flux and is denoted by the subscript "leak." The flux density distribution is given by

$$B_{\text{leak}}(\varphi) = \hat{B}_{\text{leak}} \cos(p_{\text{m}}\varphi) \tag{1}$$

where $\hat{B}_{\text{leak}}$ is the amplitude of flux density, p_{m} is the number of pole pairs of the rotor, and φ is the spatial position angle.

The relationship between the amplitude of flux density and the amplitude of flux linkage can be expressed as

$$\hat{B}_{\text{leak}} = \frac{p_{\text{m}}\hat{\Phi}_{\text{leak}}}{2rH} = \frac{p_{\text{m}}\hat{\Psi}_{\text{leak}}}{2rHN_{\text{a}}} \tag{2}$$

where $\hat{\Phi}_{\text{leak}}$ is the amplitude of the leakage flux, $\hat{\Psi}_{\text{leak}}$ is the amplitude of flux linkage, r is the air-gap radius, H is the rotor height, and N_{a} is the number of turns in the axial-force winding. The axial-force winding (denoted "a") is chosen for its larger measurement range and smaller harmonic effects compared to the drive winding.

The flux amplitude remains constant under fixed axial displacement. In the dq coordinate system, the flux of the rotating rotor can be represented as a constant, specifically, the d-axis flux linkage equals the amplitude: $\hat{\Psi}_{\text{leak}} = \Psi_{d,\text{leak}}$. For simplicity, both are denoted by Ψ_{leak}.

When the rotor undergoes axial displacement, the air-gap flux density changes, varying the flux linkage. This relationship can be used for axial position prediction and is approximately linear within a certain range [12], [36]

$$\Psi_{\text{leak}} = k_{\text{v}} z + \Psi_{\text{leak0}}. \tag{3}$$

Ideal conditions assume constant Ψ_{leak}, but in practice, factors, such as slot effects and sensor noise, introduce harmonics. The flux linkage can be expressed as a sum of spatial harmonics

$$\Psi_{\text{leak}}(\theta) \approx \sum_{j=0}^{N} \hat{\Psi}_{\text{leak},(j)} \cos\left(j\theta + \varphi_{\text{leak},(j)}\right) \tag{4}$$

where j denotes the harmonic order, $\hat{\Psi}_{\text{leak},(j)}$ is the amplitude of the (j)th harmonic, and $\varphi_{\text{leak},(j)}$ is the phase.

Current sensor noise has the most significant impact on estimation accuracy. Zero drift in phase currents introduces harmonics in the dq coordinate system after Park transformation, causing oscillations in the identified flux linkage at the electrical frequency. These appear as fundamental harmonic noise, reducing identification accuracy. As shown in (5), when the $\alpha\beta$-axis currents containing zero drift e_α and e_β are transformed via the Park matrix, the resulting d-axis current includes an additional term $e_\alpha \cos\theta + e_\beta \sin\theta$, while the q-axis current contains $e_\beta \cos\theta - e_\alpha \sin\theta$, and $\hat{i}_{\text{s}}$ is the current of winding. These terms represent harmonic components that vary synchronously

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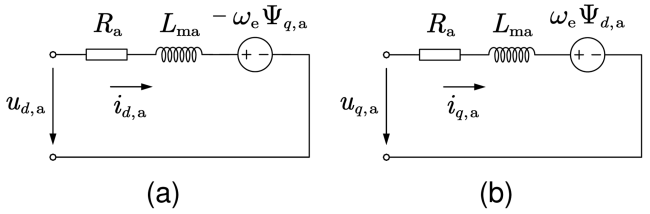


Fig. 5. Equivalent circuit diagram of the dq-axis voltage for axial permanent magnet drive system. (a) d-axis. (b) q-axis.

with the electrical angle θ, leading to periodic disturbances in both axes

$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \hat{i}_s\cos\theta + e_\alpha \\ \hat{i}_s\sin\theta + e_\beta \end{bmatrix} = \begin{bmatrix} \hat{i}_s + e_\alpha\cos\theta + e_\beta\sin\theta \\ e_\beta\cos\theta - e_\alpha\sin\theta \end{bmatrix}. \tag{5}$$

In the studied structure with eight stator teeth and concentrated windings, harmonics are inevitable. The drive and axial-force windings, both two-phase with one pole pair, exhibit fourth-order harmonics. Nonlinear rotor vibrations may also cause periodic flux linkage fluctuations [37], treated as additional harmonics. An appropriately chosen forgetting factor can filter out some of this noise, improving the signal-to-noise ratio (SNR).

Considering core magnetic saturation, permeability varies with field intensity even in the unsaturated region, altering flux linkage and inductance [38]. A nonlinear model can be established, and high-order polynomials have higher accuracy compared to linear approximations, which is

$$\Psi_{\text{leak}} = k_a i_{d,a}^2 + k_b i_{d,a} + \Psi_{\text{leak0}} \tag{6}$$

where k_a and k_b represents the fitting coefficient. And linear approximation is often sufficient. Simulation results verify that the rate of change in flux linkage remains within 0.7‰, indicating that this effect should be considered in high-precision control applications.

To achieve as accurate an estimation of the flux linkage value as possible, the voltage equation incorporating the flux linkage needs to be precisely modeled. Specifically, the stator voltage equation for the axial-force winding in the synchronous rotating reference frame can be expressed as follows:

$$\begin{cases} u_{d,a} = R_a i_{d,a} + L_{ma}\frac{di_{d,a}}{dt} - \omega_e\Psi_{q,a} \\ u_{q,a} = R_a i_{q,a} + L_{ma}\frac{di_{q,a}}{dt} + \omega_e\Psi_{d,a} \end{cases} \tag{7}$$

where $i_{d,a}$ and $i_{q,a}$ are the dq-axis currents of the axial-force winding, $u_{d,a}$ and $u_{q,a}$ are the dq-axis voltages, $\Psi_{d,a}$ and $\Psi_{q,a}$ are the dq-axis flux linkages of the stator. L_{ma} is the inductance of the single-phase axial-force winding, R_a is the resistance of the axial-force winding. ω_e is the electrical angular velocity. The equivalent circuit is shown in Fig. 5.

As shown in Fig. 6, the axial-force winding and the drive winding share the same physical phase and pole-pair number, resulting in no coupling from positional deflection angle. However, due to magnetic flux following different paths in 3-D space, certain interwinding coupling persists. Neglecting inductance coupling caused by rotor eccentricity, the dq-axis flux linkages

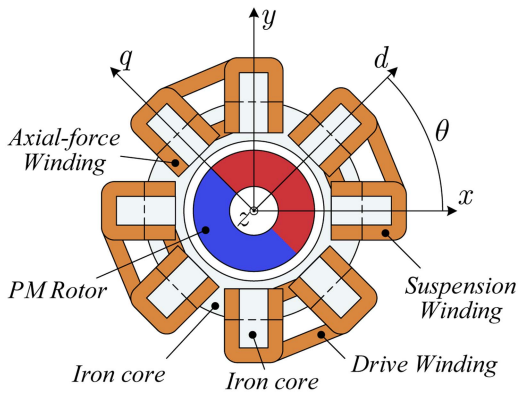


Fig. 6. Relative position of each set of windings.

for both windings can be expressed as follows:

$$\begin{bmatrix} \Psi_{d,a} \\ \Psi_{q,a} \end{bmatrix} = \begin{bmatrix} L_{ma} & 0 & M_{ad} & 0 \\ 0 & L_{ma} & 0 & M_{ad} \end{bmatrix} \begin{bmatrix} i_{d,a} + i_f \\ i_{q,a} \\ i_{d,d} \\ i_{q,d} \end{bmatrix} \tag{8}$$

where M_{ad} is the mutual inductance coefficient between the axial-force winding and the drive winding, i_f is the equivalent current of the flux linkage Ψ_{leak}, and $\Psi_{\text{leak}} = L_{ma}i_f$, and $i_{d,d}$ and $i_{q,d}$ are the dq-axis currents of the drive winding.

With $i_{d,d}$ set to 0 in field-oriented control, the dq-axis voltage equations simplify to

$$\begin{cases} u_{d,a} = R_a i_{d,a} + L_{ma}\frac{di_{d,a}}{dt} - \omega_e L_{ma} i_{q,a} - \omega_e M_{ad} i_{q,d} \\ u_{q,a} = R_a i_{q,a} + L_{ma}\frac{di_{q,a}}{dt} + \omega_e L_{ma} i_{d,a} + \omega_e\Psi_{\text{leak}}. \end{cases} \tag{9}$$

The identification expression is

$$\Psi_{\text{leak}} = (u_{q,a} - R_a i_{q,a} - L_{ma}\frac{di_{q,a}}{dt} - \omega_e L_{ma} i_{d,a})/\omega_e. \tag{10}$$

Based on (10), the flux linkage value can be estimated using online identification method.

B. Position Estimation Based on a VFF-RLS for Online Identification

The least squares identification is to determine the coefficients of a polynomial using input and output data sequences. The above equation can be written in the form of the least squares algorithm as

$$y(k) = \boldsymbol{\varphi}^{\mathrm{T}}(k)\boldsymbol{\theta} + \xi(k) \tag{11}$$

where $\boldsymbol{\varphi}(k)$ is the input data vector, $y(k)$ is the output, $\boldsymbol{\theta}$ is the vector of model parameters to be estimated, and $\xi(k)$ is the residual. The data length of $\boldsymbol{\varphi}(k)$ and $\boldsymbol{\theta}$ is l, where $\boldsymbol{\varphi}(k) = [\varphi_1(k), \varphi_2(k), \ldots, \varphi_l(k)]^T$ and $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_l]^T$.

Using the input and output data, construct the following function and minimize it:

$$J = \sum_{k=1}^{N} \lambda_{\text{RLS}}^{N-k}[y(k) - \boldsymbol{\varphi}^{\mathrm{T}}(k)\boldsymbol{\theta}]^2. \tag{12}$$

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The estimated value $\bar{\boldsymbol{\theta}}$ of $\boldsymbol{\theta}$ that minimizes J is called the least squares estimate of the parameter $\boldsymbol{\theta}$. Based on this, the expression form of RLS method with forgetting factor can be derived as follows:

$$\begin{cases} \bar{\boldsymbol{\theta}}(k) = \bar{\boldsymbol{\theta}}(k-1) + \boldsymbol{K}(k)[y(k) - \boldsymbol{\varphi}^{\mathrm{T}}(k)\bar{\boldsymbol{\theta}}(k-1)] \\ \boldsymbol{K}(k) = \boldsymbol{P}(k-1)\boldsymbol{\varphi}(k)[\lambda_{\mathrm{RLS}} + \boldsymbol{\varphi}^{\mathrm{T}}(k)\boldsymbol{P}(k-1)\boldsymbol{\varphi}(k)]^{-1} \\ \boldsymbol{P}(k) = 1/\lambda_{\mathrm{RLS}}\left[I - \boldsymbol{K}(k)\boldsymbol{\varphi}^{\mathrm{T}}(k)\right]\boldsymbol{P}(k-1). \end{cases} \tag{13}$$

The estimated value $\bar{\boldsymbol{\theta}}(k)$ at the kth instance is the previous estimated value $\bar{\boldsymbol{\theta}}(k-1)$ plus a correction term. $\boldsymbol{K}(k)$ represents the algorithm gain. $\boldsymbol{P}(k)$ represents the covariance matrix. The initial time $\boldsymbol{P}(0)$ can be set as a scalar matrix. To avoid the issue of singularity in identification, the parameter vector to be identified here is 1-D. That is, $\bar{\boldsymbol{\theta}}(k) = [\bar{\Psi}_{\mathrm{leak}}(k)]$, $\boldsymbol{\varphi}^{\mathrm{T}}(k) = [\omega_{\mathrm{e}}(k)]$, and $y(k) = u_{q,\mathrm{a}} - \omega_{\mathrm{e}} L_{\mathrm{ma}} i_{d,\mathrm{a}} - R_{\mathrm{a}} i_{q,\mathrm{a}}$.

For the forgetting factor in the RLS method, increasing λ_{RLS} enhances the averaging effect in estimation and reduces noise interference. Conversely, reducing λ_{RLS} can lead to faster dynamic performance. Using T_{s} as the sampling time, the bandwidth of estimation value can be expressed as

$$\alpha_{\mathrm{RLS}} = \frac{1 - \lambda_{\mathrm{RLS}}}{T_{\mathrm{s}}}. \tag{14}$$

To account for speed-dependent identification accuracy, a variable forgetting factor is used

$$\lambda_{\mathrm{RLS}}(n_{\mathrm{m}}) = \xi_1 e^{-n_{\mathrm{m}}^2/\xi_2} + \xi_3. \tag{15}$$

In this work, the coefficients are set as $\xi_1 = 0.001$, $\xi_2 = 10^7$, and $\xi_3 = 0.999$, respectively.

The stability of the VFF-RLS estimator is guaranteed by the boundedness of the forgetting factor. A sufficient condition for convergence is that λ_{RLS} has a positive lower bound. In our design, λ_{RLS} is strictly bounded within $[\xi_3, \xi_1 + \xi_3)$. By setting $\xi_3 = 0.999$ and $\xi_1 < 0.001$, $\lambda_{\mathrm{RLS}} \in [0.999, 1)$ is ensured. This bound prevents the covariance matrix from diverging, guaranteeing that the estimation error is uniformly ultimately bounded even during rapid speed variations.

While the lower bound $\xi_3 = 0.999$ ensures stability, the selection of $\xi_1 = 0.001$ is optimized for the SNR at low speeds. At low speeds, a higher forgetting factor is necessary to suppress noise. The exponential term governed by ξ_1 allows the forgetting factor to remain relatively high at low speeds, enhancing identification accuracy without compromising the stability margin established by ξ_3. It is observed that the accuracy of $\bar{\Psi}_{\mathrm{leak}}(k)$ is relatively insensitive to speeds above 3000 r/min, where the accuracy is acceptable for feedback. Therefore, $\xi_2 = 10^7$ is selected to maintain a relatively large forgetting factor within 3000 r/min.

Combining with (3), the rotor displacement can be estimated in real time. Furthermore, the algorithm's complexity is O(1) due to the single-parameter identification, making it computationally and readily deployable on embedded hardware.

C. Dynamic Analysis of the Mechatronic System

Choose the negative z-axis direction as the positive direction for rotor displacement and force. The forces on the rotor during stable levitation, including passive magnetic reluctance force, viscous resistance, and active axial force, can be modeled as equivalent to a spring, damper, and actuator, respectively, as shown in Fig. 7.

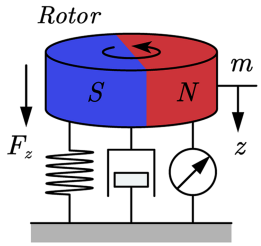


Fig. 7. Equivalent schematic diagram of mass-spring-damping-actuator for axial active-passive electromagnetic system.

When gravity is balanced by the passive levitation force, it can be omitted from the analysis. The axial dynamic equation of the rotor can be expressed in state-space form as

$$\begin{cases} \frac{\mathrm{d}z}{\mathrm{d}t} = v_z \\ \frac{\mathrm{d}v_z}{\mathrm{d}t} = \frac{k_{iz}}{m} i_{d,\mathrm{a}} - \frac{d_{\mathrm{a}}}{m} v_z - \frac{k_z}{m} z - \frac{F_{\mathrm{dz}}}{m} \end{cases} \tag{16}$$

where m is the rotor mass, z is the axial displacement, v_z is axial velocity, k_{iz} is the force-current coefficient of the axial active force, and F_{dz} is the external disturbance force. d_{a} is the axial damping coefficient. Its value can be calculated from oscillation decay, approaches zero, and can also be treated as a disturbance. The damping coefficient in water is larger than that in air, which is beneficial to stability. k_z is the force-displacement coefficient of the axial passive force, obtained from simulation.

In the application scenario, buoyancy acts on the impeller, and the net force (gravity minus buoyancy) is balanced by passive magnetic force in static suspension. This differs from the test setup, where gravity alone is balanced. While buoyancy does not affect the equilibrium, it raises the rotor's initial position, necessitating recalibration of the force-current and flux-linkage-position parameters.

Since k_{iz} is proportional to Ψ_{leak} [13], and Ψ_{leak} is proportional to z from (3), their combined relationship is

$$k_{iz} = k_{\mathrm{t}} \Psi_{\mathrm{leak}} = k_{\mathrm{t}}(k_{\mathrm{v}} z + \Psi_{\mathrm{leak0}}) = k_{\mathrm{t}} k_{\mathrm{v}} z + k_{iz0}. \tag{17}$$

Here, k_{t} is the rate of change of the force-current coefficient with respect to flux linkage, and k_{v} is the rate of change of flux linkage with respect to displacement. This relationship allows real-time approximation of k_{iz} using the current axial position z.

D. Design and Stability Analysis of an Integral SMC

This study employs SMC to address system uncertainties and external disturbances. For the mechanical subsystem, the state variables are chosen as $x = [x_1, x_2]^{\mathrm{T}} = [z, v_z]^{\mathrm{T}}$. The state-space representation derived from (16) is shown in the following:

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k_z}{m} & -\frac{d_{\mathrm{a}}}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{k_{iz}}{m} \end{bmatrix} u + \begin{bmatrix} 0 \\ -\frac{F_{\mathrm{dz}}}{m} \end{bmatrix} \tag{18}$$

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where the control input $u = i_{d,\text{ref}}$ is the reference current for the axial-force winding.

Define the position error as $e_z = z_{\text{ref}} - x_1$. To achieve zero steady-state error and improve dynamic response, an integral sliding surface is designed

$$s = c_i \int_0^t e_z \mathrm{d}\tau + c_d \dot{e}_z - c_p e_z$$
$$= c_i \int_0^t (z_{\text{ref}} - x_1)\mathrm{d}\tau + c_d(\dot{z}_{\text{ref}} - x_2) + c_p(x_1 - z_{\text{ref}}) \quad (19)$$

where c_i, c_d, and c_p are the parameters for the integral sliding mode surface, respectively. Taking the derivative of the equation yields

$$\dot{s} = c_i(z_{\text{ref}} - x_1) - c_d \dot{x}_2 + c_p x_2. \quad (20)$$

Using the exponential reaching law $\dot{s} = -\varepsilon \cdot \text{sat}(s) - qs$ (where ε and q are positive gains), and substituting (20) and the dynamics from (18), the control law is derived as

$$u = \frac{m}{c_d k_{iz}} \cdot [\varepsilon \cdot \text{sat}(s) + q \cdot s$$
$$+ c_p x_2 + c_i(z_{\text{ref}} - x_1) + c_d(\frac{d_a}{m} x_2 + \frac{k_z}{m} x_1)]. \quad (21)$$

In practice, k_{iz} varies with displacement as given in (17). For sensorless control, k_{iz} is approximated by its nominal value k_{iz0} to avoid phase lag from flux linkage estimation.

The stability of the SMC method is analyzed using the Lyapunov method [39], [40]. According to (17) and (18), the residual term introduced by linearization, parameter variations, and disturbances can be aggregated into a bounded term

$$d(t) = \frac{k_t k_v z}{m} i_{d,\text{ref}} - \frac{F_{dz}}{m} + \xi(t) \quad (22)$$

where $\xi(t)$ includes the effects of hysteresis, nonlinear coupling, and noise. The lumped disturbance $d(t)$ is bounded. $|d(t)| \leq \eta_{\text{bnd}}$.

Consider the Lyapunov function $V(s) = \frac{1}{2}s^2$. Its derivative is

$$\dot{V}(s) = s[-\varepsilon \cdot \text{sat}(s) - q \cdot s - c_d \cdot d(t)] \leq 0. \quad (23)$$

From the above equation, it can be seen that when $s > 0$, it is required to satisfy $-\varepsilon \cdot \text{sat}(s) - q \cdot s - c_d \cdot d(t) \leq 0$. Under the assumption that ε, q, and c_d are all greater than 0, the proposed method is stable only when $-\varepsilon \cdot \text{sat}(s) - q \cdot s - c_d \cdot d(t) \leq -\varepsilon \cdot \text{sat}(s) - q \cdot s + c_d \cdot \eta_{\text{bnd}} \leq 0$. Therefore, $0 < c_d \leq (\text{sat}(s) + q \cdot s)/\eta_{\text{bnd}}$. When $s = 0$, $s \cdot \dot{s} = 0$. When $s < 0$, it is required to satisfy $-\varepsilon \cdot \text{sat}(s) - q \cdot s - c_d \cdot d(t) \geq 0$. Under the assumption that ε, q, and c_d are all greater than 0, the proposed method is stable only when $-\varepsilon \cdot \text{sat}(s) - q \cdot s - c_d \cdot d(t) \geq -\varepsilon \cdot \text{sat}(s) - q \cdot s - c_d \cdot \eta_{\text{bnd}} \geq 0$. Therefore, $0 < c_d \leq (-\text{sat}(s) - q \cdot s)/\eta_{\text{bnd}}$. In summary, when satisfying $\varepsilon, q > 0$, $0 < c_d \leq (|\text{sat}(s)| + |q \cdot s|)/\eta_{\text{bnd}}$, the SMC in the position loop is stable.

IV. Simulation Results

Finite element simulations using JMAG Designer were conducted to analyze the suspension force under both steady-state and transient conditions. The model used an element size of 0.75 mm for the air-gap, 1.5 mm for parts, and 3 mm for the air region. The rotor material was Nd-Fe-B N38 and the stator was 20JNEH1200 silicon steel, consistent with the prototype. A 3-D model was built and validated against experimental mechanical parameters [38]. In addition, control algorithm simulations are performed in Simulink, where the reference frequency is set to 100 kHz to minimize the interference of discretization error on the evaluation of algorithm performance.

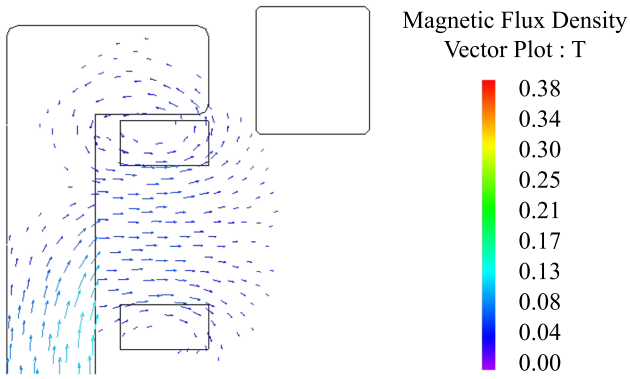


Fig. 8. Flux vector generated only by the axial-force winding in xz cross-section of 3-D simulation model.

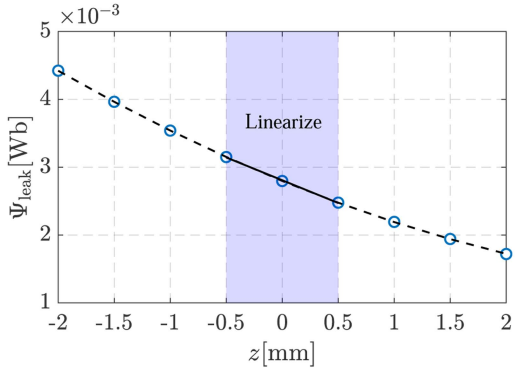


Fig. 9. Applicable boundary of linear relationship between flux linkage and axial position.

Fig. 8 shows the xz cross-section of the magnetic flux density generated by a 4 A d-axis current in the axial-force winding, with the permanent magnet flux omitted for clarity. The rotor is positioned above the winding. As it moves axially, the flux density on the rotor side varies, thereby altering the force–current coefficient.

To investigate the applicable boundary of the linear relationship between flux linkage and position, this article performs a parametric sweep based on JMAG finite element simulation. Fig. 9 shows the variation of the permanent magnet flux linkage within the extreme motion range of ±2 mm. The results indicate that a good linear relationship exists between flux linkage and position within the control range of ±0.5 mm. To further improve the accuracy of position estimation, a high-order polynomial can be used to fit the actual mapping relationship.

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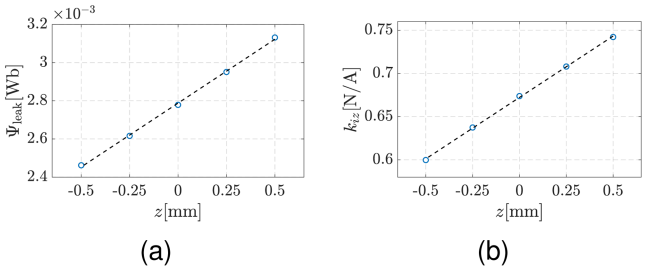


Fig. 10. Relationships between axial position and electromagnetic parameters. (a) Axial position and leakage flux linkage. (b) Axial position and force-current coefficient.

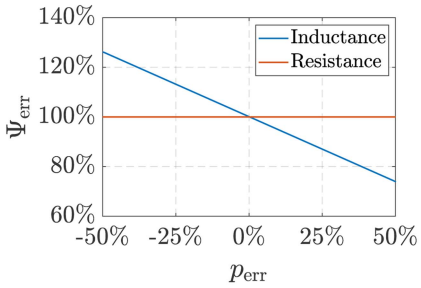


Fig. 11. Influence of system parameter (L_{ma}, R_{a}) perturbations on flux linkage identification error.

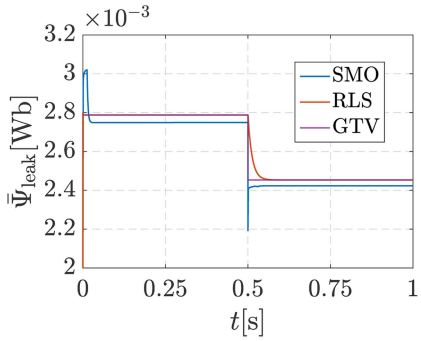


Fig. 12. Comparison of flux linkage obtained through a SMO and least squares identification in Simulink.

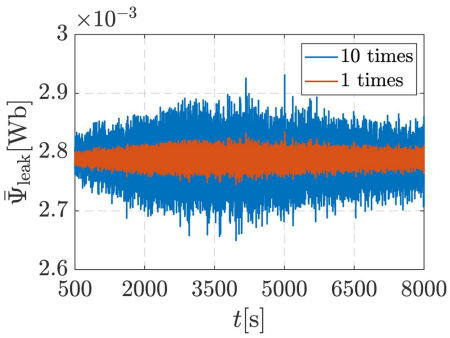


Fig. 13. Influence of noise power on flux linkage identification error.

Transient simulations conducted at various rotor axial positions reveal a strong linear relationship between leakage flux linkage and axial displacement within the range of -0.5 to 0.5 mm, as shown in Fig. 10(a). The relationship is defined by $\Psi_{\mathrm{leak}} = 0.0006696z + 0.002788$ ($R^2 = 0.9990$), allowing reliable displacement estimation from flux linkage. According to (17), this variation in flux linkage directly influences the force-current coefficient k_{iz}, leading to a similarly linear behavior between k_{iz} and z. Steady-state simulation results in Fig. 10(b) confirm this relationship, which is expressed as $k_{iz} = 0.1421z + 0.6722$ ($R^2 = 0.9995$).

To investigate the flux linkage identification error caused by motor parameter errors, a sensitivity analysis is conducted. In the simulation, the inductance and resistance are adjusted separately to examine their effects on the identification results. The current component $i_{d,a} = 1$ A is set to emulate actual rotor driving conditions. The results in Fig. 11 show that inductance variations in the range of (-50%) to 50% lead to flux linkage identification errors of (-26%) to 26%, while resistance variations in the same range cause no observable identification error. This phenomenon is attributed to the setting $i_{q,\mathrm{a}} = 0$ in vector control, which eliminates the resistance term from the dq-axis voltage equations, making the system insensitive to resistance errors. Fig. 12 compares the sliding mode observer (SMO) observation method based on phase windings and the RLS identification method under ideal simulation conditions. The forgetting factor corresponding to the speed of 4000 r/min was 0.9992. At this time, the position calculation bandwidth was 80 Hz. A 0.5 mm dynamic displacement is introduced starting at the 0.5-s mark to force a change in flux linkage. The flux linkage identified dynamically by the RLS algorithm tracks the ground truth value accurately, with zero steady-state error. However, the flux linkage calculated through the SMO based on phase windings exhibits errors in steady state, and these errors also vary with the speed. Due to the lack of closed-loop flux amplitude regulation in the traditional SMO, its output remains unstable. Thus, the RLS method shows significant advantages over the SMO.

To verify the performance of the proposed method under "worst-case" conditions, the simulation sets the speed to accelerate from 500 to 8000 r/min with Gaussian noise at ten times the reference noise power. As shown in Fig. 13, under the constraint of the variable forgetting factor function, the identified flux linkage maintains high accuracy at low speeds and remains stable throughout the entire speed range without divergence, confirming the convergence of the method. Compared with the fixed forgetting factor approach, the proposed VFF-RLS method exhibits superior stability and accuracy under extreme conditions.

V. Experiment and Test Results

A. Hardware Equipment

The software layout and hardware diagram of the control system are shown in Figs. 14 and 15, respectively, with the latter also illustrating the hardware structure and dynamic suspension control platform. The controller is a dSPACE DS1103, where the

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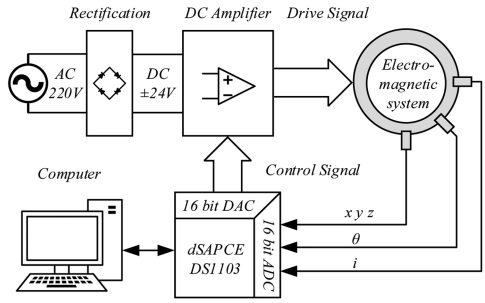


Fig. 14. Layout diagram of control system software.

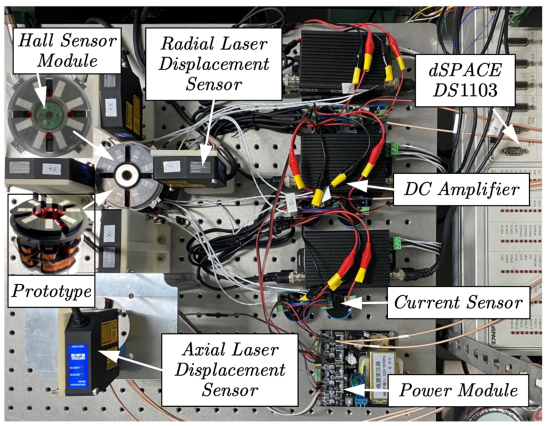


Fig. 15. Experimental platform of the electromagnetic system.

TABLE I
PARAMETERS FOR PROTOTYPE MEASUREMENT AND TESTING

Symbol	Quantity	Value
m	Rotor Mass	0.045kg
R_{a}	Resistance of axial-force winding	5.315Ω
L_{ma}	Measured inductance	1.458mH
$L_{\mathrm{ma,corr}}$	Corrected inductance	1.924mH
M_{ad}	Mutual inductance coefficient	0.379mH
k_{iz}	Force-current coefficient	$0.60-0.74$N/A
k_{iz0}	Initial Force-current coefficient	0.67N/A
k_z	Force-displacement coefficient	4.02N/mm
c_p	Parameter of sliding surface	8.77
c_i	Parameter of sliding surface	80.80
c_d	Parameter of sliding surface	1
ε	Parameter to overcome disturbances	0.1
q	Parameter for approaching	9.21

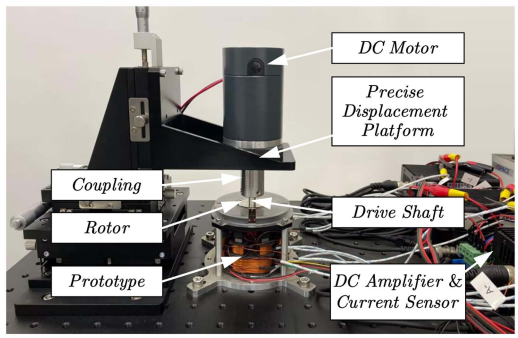


Fig. 16. Experimental platform for flux linkage testing.

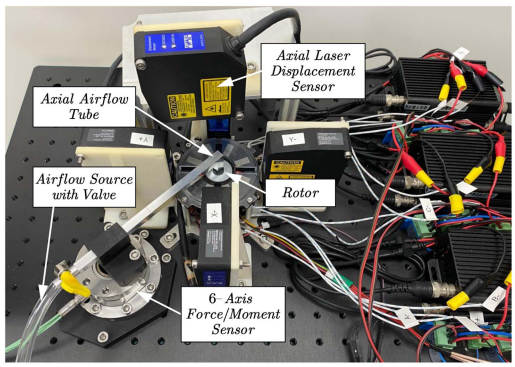


Fig. 17. External disturbance loading platform.

PowerPC serves as the main processor for overall control and complex computations, while the DSP acts as a coprocessor for high-speed concurrent digital signal processing. This platform enables rapid control prototyping based on real-time simulation to validate control algorithms. For signal acquisition, the system uses analog-to-digital converter (ADC) to receive voltage analog signals from displacement sensors, angle sensors, and current sensors. Specifically, radial and axial displacement are measured by OPTEX FA CD5-30 A and CD5-85 laser sensors, respectively, while the rotor angle is detected by a Hall sensor module. Control signals are output to the power amplifier via the bipolar digital-to-analog converter (DAC) module of the dSPACE DS1103. The power amplifier used is a FeelElec FPA101 A bidirectional DC linear power amplifier. These components are labeled in the figures. The parameters of the test platform are given in Table I.

To verify the feasibility of the proposed algorithm in embedded hardware, the reference sampling time is set to $T_{\mathrm{s}} = 10^{-4}$ s, resulting in a control frequency of 10 kHz. The laser displacement sensor used has a maximum sampling frequency of 10 kHz; the linear Hall sensor (for rotor angle measurement) has an operating bandwidth of 17 kHz; the current sensor has a measurement bandwidth of 100 kHz; the linear power amplifier can achieve a sinusoidal output with a bandwidth of 100 kHz within full power range.

Fig. 16 illustrates the flux linkage testing platform. The rotor is fixed to the external rotating motor via the coupling, and the motor is mounted on the precise displacement platform. By adjusting the axial displacement, the flux linkage varies, causing changes in the currents detected by the current sensors, which in turn leads to variations in the identified flux linkage.

Fig. 17 shows the axial disturbance platform. The axial disturbance force on the rotor is generated by high-speed airflow impacting its surface. The axial airflow tube is fixed to a six-axis force/torque sensor to measure the reaction force, which is used to approximate the force acting on the rotor. The airflow source is connected to a flow control valve for adjusting the airflow rate, thereby regulating the disturbance force magnitude. The airflow disturbance is employed solely for qualitative validation of the controller's disturbance rejection capability, and thus differs

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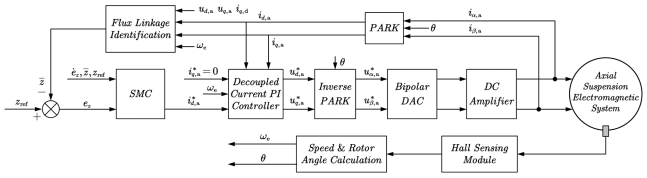


Fig. 18. Axial position control diagram with SMC.

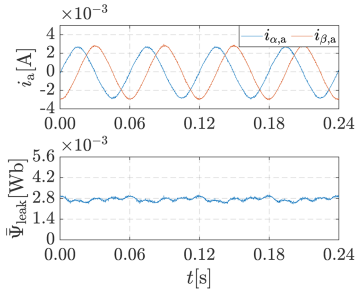


Fig. 19. Current of phase α and β of the axial-force winding under external drive at 1000 r/min and the converted flux linkage.

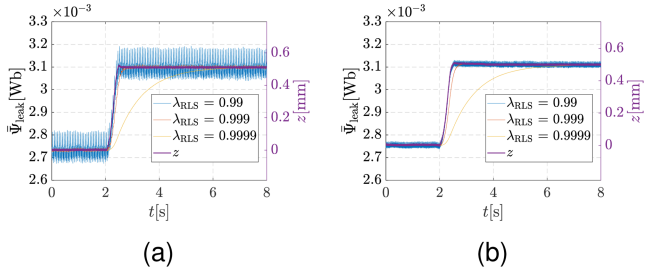


Fig. 20. Identified flux-linkage values of the flux linkage by setting different forgetting factors. (a) 1000 r/min. (b)3000 r/min.

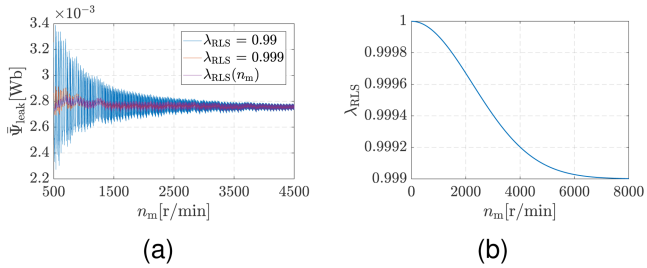


Fig. 21. Comparison of identification performance and adaptive forgetting factor tuning. (a) Identified flux-linkage values at different rotational speeds and forgetting factors. (b) Curve of the forgetting factor set by (15).

from actual hydraulic loads. Despite this, it has the advantages of convenient loading, good repeatability, and adjustable force magnitude.

B. Control Method

As shown in Fig. 18, the magnetic field orientation control used here consists of a current inner loop with a PI controller and a position outer loop using the proposed integral-based SMC. A RLS method is applied for online flux linkage identification. The computed position $\bar{z}$ provides feedback to the outer loop.

The proportional and damping coefficients were initially determined via pole placement to achieve the desired equivalent stiffness and damping of the system [28]. These intermediate coefficients of the linearized system are then transformed into the specific values of c_p, c_i, and c_d, as summarized in Table I for the SMC parameters.

C. Experiment Results

As shown in Fig. 19, the amplitude and phase of currents in the two-phase windings and the resulting flux linkage are presented, with the rotor externally driven at 1000 r/min (via the test bench in Fig. 16). The flux linkage here is directly calculated from the voltage (9), but results indicate slight fluctuations in the computed flux. Therefore, this calls for the use of the RLS method. The average flux linkage is about 0.002731 Wb, which aligns with the simulation.

Fig. 20(a) and (b) shows the identified flux-linkage values at 1000 and 3000 r/min under different forgetting factors. During testing, the rotor was externally driven while its axial displacement was manually adjusted from 0 to 0.5 mm using a translation stage. Results show that with $\lambda_{\mathrm{RLS}} = 0.99$, the identified value exhibits almost no phase lag compared to the displacement sensor measurement. A higher forgetting factor improves precision but introduces time delay. At lower speeds, the SNR is low, causing larger fluctuations in flux estimates. At higher speeds, system precision is inherently better, allowing a lower forgetting factor to improve convergence. Thus, the forgetting factor should be adjusted dynamically to balance precision and convergence speed across different rotational speeds.

As shown in Fig. 21(a), the identified magnetic flux value is highly sensitive to rotational speed, with significant fluctuations occurring particularly at lower speeds. (Speeds below 1000 r/min are seldom used in practice.) To address this, a function given by (15) has been developed to dynamically adjust the forgetting factor according to rotor speed, thereby improving low-speed position estimation accuracy. As illustrated in Fig. 21(b), λ_{RLS} varies between 0.999 and 1 based on the rotational speed. Therefore, a variable forgetting factor scheme is adopted to adjust identification accuracy according to speed, thereby improving both low-speed precision and high-speed bandwidth.

Fig. 22(a) and (b) presents the correlation between the identified flux linkage Ψ_{leak} and the axial position z of the rotor in both simulation and experiment. In the Simulink simulation, the axial position was adjusted via the d-axis current, while in the experiment, a precision displacement platform was used. In both cases, the flux linkage was identified in real-time using voltage equation parameters. The results show a strong linear relationship between Ψ_{leak} and z within the operational range. Linear fitting yields $\Psi_{\mathrm{leak}} = 0.0006696z + 0.002788$ for simulation and $\Psi_{\mathrm{leak}} = 0.0006855z + 0.002758$ for experiment. The experimental result is consistent with simulation and will serve as the calibration value in actual control. During rotor rotation

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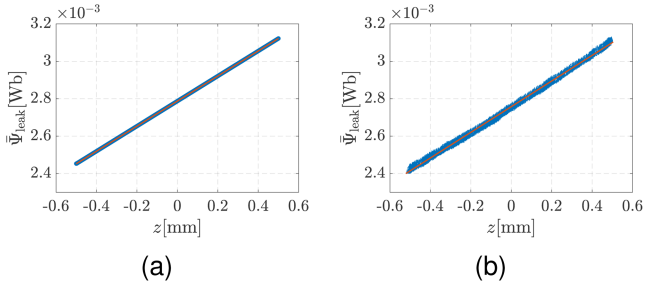


Fig. 22. Relationship between flux linkage and axial displacement. (a) Simulation. (b) Experimental.

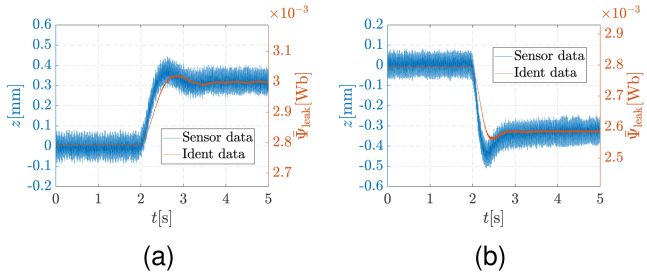


Fig. 23. Sensorless control of axial displacement. (a) Positive direction of the rotor. (b) Negative direction of the rotor.

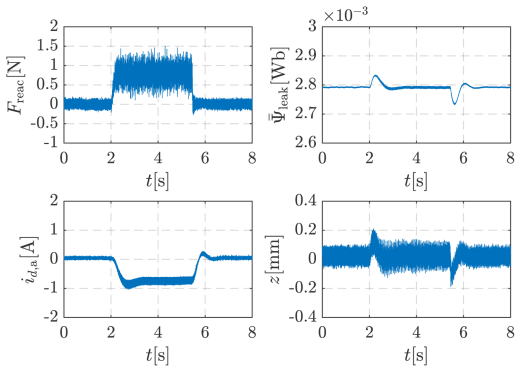


Fig. 24. Active control of rotor under axial airflow disturbance at 4000 r/min.

and suspension, the simulation based method for calibrating flux linkage and position was applied experimentally by adjusting rotor position through current drive. The experimental results indicated a higher rate of change of Ψ_{leak} with respect to z compared to simulations. This discrepancy is attributed primarily to an underestimation of the inductance L_{ma}, which leads to an overestimation of the identified flux linkage. According to (9), if L_{ma} in the term $L_{\text{ma}} i_{d,\text{a}}$ is smaller than the actual value, the identified leakage flux Ψ_{leak} exhibits dependence on current. This behavior was confirmed through tests in an idealized Simulink environment.

To resolve this issue, the inductance value must be corrected to eliminate the influence of $i_{d,\text{a}}$ variations on Ψ_{leak}. The corrected parameter $L_{\text{ma,corr}}$ should then be used in the identification model.

To further analyze how L_{ma} affects Ψ_{leak}, the partial derivative of (9) with respect to $i_{d,\text{a}}$ is considered as

$$\frac{\partial \Psi_{\text{leak}}}{\partial i_{d,\text{a}}} = -L_{\text{ma}} + \frac{\partial}{\partial i_{d,\text{a}}} \left(\frac{u_{q,\text{a}} - R_{\text{a}} i_{q,\text{a}}}{\omega_{\text{e}}} \right). \tag{24}$$

This expression shows that a smaller L_{ma} reduces the magnitude of the first term $(-L_{\text{ma}})$, thereby increasing the absolute value of $\frac{\partial \Psi_{\text{leak}}}{\partial i_{d,\text{a}}}$. As a result, Ψ_{leak} exhibits greater sensitivity to changes in $i_{d,\text{a}}$.

Owing to the absence of a magnetic core, the quality factor of the axial-force winding is relatively low. For such windings, the inductance values obtained via inductance-capacitance-resistance (LCR) measurement exhibit significant discrepancies depending on the choice of series or parallel equivalent circuit models, which introduces errors into model-based identification. Consequently, dynamic identification and initial calibration of the axial-force winding are required under rotor suspension and rotation. The resistance and inductance values obtained from LCR measurements, together with the corrected inductance values, are presented in Table I.

As shown in Fig. 23, the rotor was commanded to track a dynamic axial displacement of ± 0.3 mm. The identified flux linkage served as the feedback signal for control, while a laser displacement sensor provided synchronous measurements for validation. The results indicate that the flux linkage accurately tracks the rotor position, enabling precise motion control. Displacement fluctuations remained within 160 μm, and the rotor achieved stable dynamic positioning within a 1 s time interval.

The ability to maintain accurate control without a dedicated position sensor confirms the effectiveness of the flux-linkage-based sensorless method.

Fig. 24 shows the sensorless axial position control performance under airflow disturbance at 4000 r/min. The rotor maintained its position near zero under approximately 3 s of disturbance. The six-axis sensor measured an average reaction force of 0.79 N, though the actual force on the rotor was lower. A current of $i_{d,\text{a}} \approx -0.74$ A resisted the displacement. The disturbance force showed high-frequency fluctuations, resulting in rotor displacement variations between 160 and 230 μm. The sensorless method achieved stable axial positioning overall. Prioritizing suspension accuracy led to high forgetting factors and low identification bandwidth, limiting high-frequency disturbance rejection. However, such high-frequency effects are expected to be less significant in high-damping environments like water.

Fig. 25 shows sensorless control performance under airflow disturbance at 1000 r/min. The rotor maintained its position near zero under approximately 12 s. The average reaction force measured is about 1.33 N and the current $i_{d,\text{a}}$ is about -1.08 A. The rotor displacement variations between 230 and 340 μm. Due to the narrowed bandwidth, the required stabilization time has also increased.

Fig. 26 shows the axial impulse response under sensorless control at 4000 r/min. A transient force of 3.164 N was applied at 3 s through a fast-acting valve. The active electromagnetic control suppressed axial vibration, restricting maximum displacement to 0.5074 mm.

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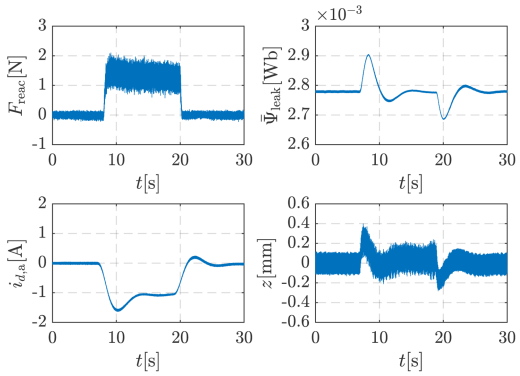


Fig. 25. Active control of rotor under axial airflow disturbance at 1000 r/min.

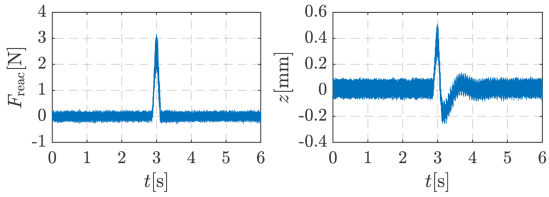


Fig. 26. Active control of rotor under axial airflow impulse disturbance.

VI. Conclusion

This study employs a RLS method to identify rotor flux linkage for axial position estimation, which serves as feedback in the position control loop. An adaptive forgetting factor is utilized to adjust the identification bandwidth and mitigate flux fluctuations at low speeds. The strong linear relationship between flux and position within the operational range enables sensorless axial control, and the closed-loop structure further provides disturbance rejection capability.

The proposed sensorless scheme integrates drive and position detection to reduce system cost and spatial requirements, making it extensible to a wide range of permanent magnet synchronous systems. However, current limitations include a lower speed bound of 1000 r/min, as well as the tradeoff between estimation precision and convergence speed due to the large forgetting factor required by limited flux linkage measurement accuracy.

Future work will address these issues by employing high-precision sensors to improve identification performance, alongside developing intelligent algorithms, nonlinear flux-position models, and advanced controllers to enhance signal quality, reduce latency, and achieve higher control accuracy. Specifically, intelligent optimization algorithms will be utilized to determine the multidimensional parameters of sliding mode surface coefficients, with the overarching priority placed on optimizing the long-term stability and reliability of the suspension drive system. Underwater experiments will also be conducted for further validation.

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